Antiderivatives and Riemann Sums

ANSWER KEY

Basic antiderivatives

1 Find the general antiderivative of each function:

a $f(x) = x^4 \quad = \frac{x^5}{5} + C$

b $f(x) = \frac{1}{x} \quad = \ln|x| + C$

c $f(x) = e^x \quad = e^x + C$

d $f(x) = \sec^2 x \quad = \tan x + C$

2 Find $\int (3x^2 - 4x + 5)\,dx$.
$= x^3 - 2x^2 + 5x + C$.

3 Find the particular antiderivative $F(x)$ of $f(x) = 6x^2 - 4$ satisfying $F(1) = 5$.
$F(x) = 2x^3 - 4x + C$. $F(1) = 2 - 4 + C = 5 \Rightarrow C = 7$. $F(x) = 2x^3 - 4x + 7$.

Riemann sums: left, right, and midpoint

4 Approximate $\int_0^4 x^2\,dx$ using a Left Riemann Sum with $n = 4$ equal subintervals.
$\Delta x = 1$; left endpoints $x = 0, 1, 2, 3$: $L_4 = 1[0^2 + 1^2 + 2^2 + 3^2] = 1[0 + 1 + 4 + 9] = 14$.

5 Approximate $\int_0^4 x^2\,dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.
$\Delta x = 1$; right endpoints $x = 1, 2, 3, 4$: $R_4 = 1[1 + 4 + 9 + 16] = 30$.

6 Approximate $\int_0^4 x^2\,dx$ using a Midpoint Riemann Sum with $n = 4$ equal subintervals.
$\Delta x = 1$; midpoints $x = 0.5, 1.5, 2.5, 3.5$: $M_4 = 1[0.25 + 2.25 + 6.25 + 12.25] = 21$. (True value: $\frac{64}{3} \approx 21.33$.)

The Trapezoidal Rule

7 Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^4 x^2\,dx$.
$T_4 = \frac{L_4 + R_4}{2} = \frac{14 + 30}{2} = 22$.

8 A table gives $f(0) = 2$, $f(2) = 5$, $f(4) = 9$, $f(6) = 8$. Use the Trapezoidal Rule with these unequal-width data (widths of 2 each) to approximate $\int_0^6 f(x)\,dx$.
With 3 equal subintervals of width 2:
$T = 2\left[\frac{f(0) + f(2)}{2} + \frac{f(2) + f(4)}{2} + \frac{f(4) + f(6)}{2}\right] = 2\left[\frac{7}{2} + \frac{14}{2} + \frac{17}{2}\right] = 7 + 14 + 17 = 38$.

Sums as limits: the definite integral

9 Explain how the definite integral $\int_a^b f(x)\,dx$ is defined as a limit of Riemann sums, and write the limit expression using sigma notation.

$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$, where $\Delta x = \frac{b-a}{n}$ and x_i^* is any sample point in the i-th subinterval — as $n \to \infty$ (subintervals shrink to zero width), all Riemann sum types (left, right, midpoint) converge to the same value, the exact signed area under f.

10 Determine whether the Left Riemann Sum overestimates or underestimates $\int_0^4 x^2\,dx$, and explain why using the shape of the graph.

Since $f(x) = x^2$ is increasing on $[0, 4]$, the left endpoint of each subinterval gives the smallest function value on that subinterval, so the Left Riemann Sum UNDERESTIMATES the true area (confirmed: $L_4 = 14 < \frac{64}{3} \approx 21.33$).

Riemann sums from a table of values

11 A table gives $f(0) = 3$, $f(1) = 5$, $f(2) = 8$, $f(3) = 10$, $f(4) = 9$. Use a Right Riemann Sum with 4 subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.

$R_4 = 1[f(1) + f(2) + f(3) + f(4)] = 5 + 8 + 10 + 9 = 32.$

Antiderivatives of trigonometric and exponential functions

12 Find $\int (3\sin x + 2e^x)\,dx$.

$= -3\cos x + 2e^x + C.$

13 Find the particular antiderivative $F(x)$ of $f(x) = \sec^2 x$ satisfying $F(0) = 2$.

$F(x) = \tan x + C$. $F(0) = 0 + C = 2 \Rightarrow C = 2$. $F(x) = \tan x + 2$.

Visualizing a Riemann sum approximation

14 The graph shows $f(x) = x^2$ on $[0, 4]$. Use the graph to explain why the Right Riemann Sum $R_4 = 30$ (found earlier) overestimates the true area, and why the Left Riemann Sum $L_4 = 14$ underestimates it.

Since $f(x) = x^2$ is increasing on $[0, 4]$, right-endpoint rectangles reach up to the curve's height at the RIGHT edge of each strip, so their tops poke above the curve, overestimating the area. Left-endpoint rectangles use the height at the LEFT edge (lower, since f is increasing), so their tops stay below the curve, underestimating the area.

An applied antiderivative problem

15 A factory's production rate is $r(t) = 100 + 20t$ units/day, where t is days since a process change. Find the total production function $P(t)$ given $P(0) = 0$, then find the total production over the first 10 days.

$P(t) = \int r(t)\,dt = 100t + 10t^2 + C$. Using $P(0) = 0$: $C = 0$. $P(t) = 100t + 10t^2$.
$P(10) = 1000 + 1000 = 2000$ units.

More antiderivative practice

16 Find the general antiderivative of $f(x) = 4x^3 - \frac{2}{x^2} + \cos x$.

$= x^4 + \frac{2}{x} + \sin x + C$.

17 Find the particular antiderivative $F(x)$ of $f(x) = x^2 - 3$ satisfying $F(0) = 4$.

$F(x) = \frac{x^3}{3} - 3x + C$. $F(0) = 0 - 0 + C = 4 \Rightarrow C = 4$. $F(x) = \frac{x^3}{3} - 3x + 4$.

Comparing Riemann sums to the exact integral

18 For $\int_0^4 x^2\,dx$, the exact value is $\frac{64}{3} \approx 21.33$. Rank the four approximations found earlier — $L_4 = 14$, $R_4 = 30$, $M_4 = 21$, $T_4 = 22$ — from closest to farthest from the exact value, and explain which type of approximation (midpoint or trapezoidal) tends to be more accurate for a smooth curving function.

Distances from 21.33: M_4 ($|21 - 21.33| = 0.33$), T_4 ($|22 - 21.33| = 0.67$), L_4 ($|14 - 21.33| = 7.33$), R_4 ($|30 - 21.33| = 8.67$). Order (closest to farthest): M_4, T_4, L_4, R_4. The midpoint approximation tends to be more accurate than the trapezoidal for smooth curves, because its errors are typically about half the size and opposite in sign to the trapezoidal error.